

Reading Textbooks and Taking Notes

Econ 113 • Fall 2026 • Discussion 01A/01B

Reading (economics) textbooks is difficult

Getting through a dense textbook can feel impossible, especially once the course gets rolling. You may ask, where should I start? What if I don't start until Week 5 - is it too late? How can I find the time to read the textbook and do the practice problems? All of these are valid questions and I have a few answers:

- 1 **Textbooks are not designed to be read cover-to-cover.** They are dense and lack a narrative arc. But, they are still extremely useful. It's never too late to start learning with the textbook.
- 2 **Use the book when you need it.** If this is your first class discussing heteroskedasticity - you're in luck. Go to Chapter 8 of the textbook, and begin reading, section-by-section. If there is a section you feel comfortable with from prior knowledge or from course notes or lecture, just skip it.
- 3 **Textbooks have incredible appendices** useful for walking through the math line by line. Check your answers against these where possible.
- 4 **Use the index!** The index is your best friend. In the back of the textbook, it can be used to find the specific pages in the textbook that cover the topic. Need help understanding the *t*-statistic? Flip to the index and see there are multiple pages referencing the t-stat, for asymptotics, in relation to the F stat, and for understanding heteroskedasticity-robust estimates
- 5 **Use the textbook to prepare for lecture or review concepts presented in lecture.** For example, either before or after Lecture #10 on OLS properties (depending on your preference for how to study) read Chapters 2.2-2.5 and 3.5. The professor even points you there in the slides.

Practical ways to get started

Read the textbook in small pieces: **5–10 pages at a time**. Take your notes **by hand**. You retain way more information when you hand write information. This process of reading, digesting and interpreting economic and statistical concepts into your own word via the act of paraphrasing is essential to true learning. As you read, work through the practice problems and worked examples in the textbook as you go.

Recommendation:¹ Get in the habit of reading the textbook and taking notes. I recommend the following: (1) using the Cornell note taking method. Feel free to use the components of the method loosely, or however you adapt to your current system of notetaking. This can be combined with elements of the CERIC approach to reading (claim, evidence, reasoning, and in some cases, the implication). And (2) glossing. Glossing is the process of writing down the words you do not understand and defining them as you go, so you learn new concepts and vocabulary (i.e. panel data, sample size, regressor, etc).

Below are resources to get you started:

- Cornell notes has a full module (public, takes about 15 minutes to complete) to learn the process and benefits of this note-taking style: <https://canvas.cornell.edu/courses/1451>
- Cornell notes one-pager overview: <https://subjectguides.york.ac.uk/note-taking/cornell>
- The CERIC Method Cheat Sheet: <https://cericmethod.com/ceric-cheat-sheet/>

¹This document is designed to support students in actively and productively reading the textbook. It's very feasible to apply this method of note-taking to lecture, as well, and I encourage you to do so.

Learning how to take good notes

What are useful notes? Shorthand notes that capture the gist of what is presented in lecture or in the textbook. Using the Cornell note-taking approach, you should set up three "areas" on your page:

Cue column After lecture or reading: questions. What do you not understand? What needs clarifying?	Main notes area Take your notes here during lecture or while reading, however you like.
Summary area Afterwards: the core takeaways from this section of your notes.	

Note-taking should not be copying. Notes are designed to incorporate short form sentences, abbreviations, images, diagrams or comments that help you make sense of the information presented. You do not have to adhere to the note-taking structure above, but its a good starting point if you feel your notes are not sufficient at helping you learn.

Why write notes by hand? Distilling facts and information into handwritten notes is how you make sense of the material, and it is how you will make connections between the concepts in this course. Without the process of paraphrasing, you fail to move from just listening or recording information, to truly understanding it in a way that you can explain it clearly to a colleague, classmate or most importantly, on an exam.

- **Notes improve recall.**
- **Asking questions** during the learning process activates higher-level thinking beyond memorizing: application and evaluation.
- **Summarizing** the notes gives you the big picture, being able to connect it back to the wider concepts of the course material, now that you have the details.

What happens after you take notes

After class or after reading the textbook (preferably the same day) take **10–20 minutes** to review your notes:

1. Re-read your notes
2. Fill in any information you missed
3. Clarify: if something is wrong, cross it out and write the correct information (in a different color possibly for clarity)
4. Where you missed something, use the textbook to find the definitions and concepts that were covered but that you could not get down during lecture
5. Cross-check your notes against that day's slides
6. Write the summary at the bottom: What are the core lines of argument? What pieces of information (definitions, models, etc.) did you learn? What concepts do they connect to?

A note on AI tools

Skipping the work of paraphrasing new material and putting it in your own words will stifle your critical thinking, leave you with low retention of the material, and prove frustrating on exams, when you cannot recall the information or apply the concepts you have seen. You will need to start getting in the habit of talking about economics

- I urgently recommend against AI note-taking apps. They are **not allowed in section** as it is an invasion of your classmates' privacy to record section without full authorization by all students. Secondly, sections are hands-on, and discussions based on questions that students bring forward, and therefore are not things you can distill via AI and usefully learn from.
- I strongly suggest you do not use AI to summarize course material or to ask for "the important concepts." You need to learn the material yourself and work out how the concepts relate to each other. You have countless resources (discussion sections, office hours with TA or professor, LSS groups, econ study hall, and the ability to form your own problem set study groups with classmates) that are far more effective ways of learning the material.

Reading for claim, evidence, reasoning, and implication

While more often applied to reading research articles, it may also be useful to use the CERI portion of the CERIC+ method to understand how the textbook presents new concepts, and supports those concepts with the theoretical underpinning. In this class, a large amount of the work will be in interpreting the statistical analysis for causality, and therefore the implications of the models, tests, and functions is that they help us build causal claims. You can consider using these steps when reading the textbook. Start by identifying (highlight, copy, or underline) each component in a section of the textbook.

Claim: the new knowledge, or the statement being made.

Evidence: the data or theory that supports the claim.

Reasoning: the logic which is normally presented in the form of an analytical approach that connects the evidence to the claim.

Implication: what the claim lets us do or understand next.

Application: Lecture 5, correlation and covariance.

Claim: Covariance is a measure of the linear dependence between two random variables.

Evidence: We use the definition of expectation, applied to random variables, to show that covariance measures dependence, specifically, the linear part of it.

Reasoning:

- Define $X - \mu_X$ (the deviation of X from its population mean), and likewise $Y - \mu_Y$. We are looking at the joint expectation of these random variables.
- Apply the expectation operator, since they are random: $\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$.
- Rearrange using the properties of expectations: $\text{Cov}(X, Y) = E(XY) - \mu_X \mu_Y$. This is the population covariance, which differs from the sample covariance.

Implication: Covariance tells us how two variables move together. This matters when we build a regression where we want to isolate the effect of one variable on another.

Summary: Covariance measures linear dependence. We measure it by treating X and Y as random samples from a population; these must behave within the rules of statistics, which lets us show what the linear dependence is actually measured by. Statistical properties must be understood to make causal claims.